

Agentic Travel Assistant – System Documentation

1. System Architecture and LangGraph Design

The system is designed as an agentic backend service using FastAPI, LangGraph, and external travel APIs. The FastAPI layer exposes a POST /query endpoint. The request is passed to a LangGraph agent which maintains state and orchestrates execution. The first node is an LLM parser that extracts intent and structured parameters (origin, destination, dates). Based on the intent, conditional edges trigger the flight search node, hotel search node, or both. These nodes call the travel provider API and return structured results. A synthesis node then uses the LLM to generate a user-friendly summary. The final response includes both structured data and natural language text.

2. Testing and Evaluation Strategy

Evaluation includes unit tests, integration tests, and agent correctness metrics. Unit tests validate tool logic and parameter validation. Integration tests simulate real user queries and verify that correct tools are triggered. Metrics include Tool Trigger Accuracy, Parameter Extraction Accuracy, Response Latency, and Hallucination Rate. Mock APIs are used in CI pipelines to ensure deterministic results. Human evaluation is also used to assess response quality.

3. Local LLM Migration Plan

The LLM client is abstracted to allow switching providers. Initially, cloud models are used. Migration involves deploying a local model using Ollama or vLLM. The `llm_client` module will call the local inference endpoint instead of the cloud API. Trade-offs include increased hardware requirements but improved privacy and reduced per-request cost. GPU acceleration is recommended for acceptable latency.

4. Design Decisions and Assumptions

FastAPI was chosen for performance and async support. LangGraph was selected for explicit agent state management. External travel APIs provide real-time data. The system is modular, allowing tools and LLM providers to be replaced easily. It assumes valid API credentials and internet connectivity. Mock responses are used in development environments.